GlucoBloom: Personalized Digital Twin for Glucose Forecasting

Digital Twin Challenge 2026 • Prototype & Code Submission

Healthcare Focus: Non-invasive glycemic forecasting, personalized metabolic adaptation, and clinical safety.

Target Users: Individuals with Type 1 & Type 2 Diabetes, endocrinologists, and clinical care teams.

Core Innovation: Moving from retrospective glucose monitoring to prospective, uncertainty-calibrated 30-minute forward forecasting with personal digital twin evolution.

Problem Statement & Clinical Healthcare Need

Why Traditional Glucose Trackers Fail

• **Retrospective Blindspot:** Commercial CGMs display where glucose was 5 minutes ago. Patients lack actionable foresight to prevent hypoglycemia (<70 mg/dL) or postprandial spikes (>180 mg/dL) before symptoms manifest.

• **Biological Non-Linearity:** Identical carbohydrate meals provoke drastically different glycemic excursions across individuals due to distinct insulin sensitivities, gastric emptying kinetics, and circadian phases.

• **Alarm Fatigue & Dangerous Over-Correction:** Static population models fail to adapt, frequently causing false alarms or misleading patients into insulin stacking.

Two-Level Digital Twin Architecture

Scalable Population Knowledge + Rapid Personal Adaptation

• **Level 1: General Foundation Model (XGBoost Regressor):** Trained across 160 diabetic patients (598,960 readings) to master universal glucose kinetics, velocity, and diurnal basals.

• **Level 2: Personalized Digital Twin Adapter:** Continued gradient boosting fine-tuning on an individual's personal history, capturing unique carb sensitivities without catastrophic forgetting.

• **Uncertainty Quantification (Quantile Regressors):** Specialized models for 10th and 90th percentiles, yielding a calibrated 80% prediction interval cone.

Empirical Results on 25 Held-Out Test Patients

Zero Patient Leakage • 93,568 Test Readings

Model / Strategy	MAE (mg/dL)	RMSE (mg/dL)	Improvement vs Baseline	Clinical Safety
Persistence Baseline (t = t+30m)	17.20	29.59	0.0% (Reference)	High lag error
Linear Regression Benchmark	10.89	15.60	+36.7%	Partial tracking
XGBoost General Foundation	7.64	11.94	+55.6%	Strong accuracy
Personalized Twin (Patient 175)	6.44	9.81	+26.4% Personalization	100.0% Zone A+B

Clinical Safety: In Clarke Error Grid Analysis (EGA), **100.0% of predictions fall in Zones A + B** (99.6% in Zone A alone), confirming zero dangerous treatment errors.

The Five Signature Innovations of GlucoBloom

Demo-Ready, Clinically Grounded Features

1. ■ **30-Minute Forecast Cone:** Real-time trajectory with calibrated 80% uncertainty bounds + CSV uploader.
2. ■ **Digital Twin Evolution:** Visual audit proving error reduction from 8.74 down to 6.44 mg/dL as personal data grows.
3. ■ **Glucose Time Machine & Clarke Grid:** Step-by-step retrospective comparison against reality with gold-standard medical validation.
4. ■ **Scenario Studio ('Fan of Futures'):** Branching 3-path simulations (Sedentary vs 15m walk vs 30m brisk walk) + Food Diary.
5. ■■ **Smart Data Quality Checker:** Sensor anomaly, dropout, and compression artifact detection to protect prediction confidence.

Ethics, Privacy & Production Architecture

Safety First • Production Ready

- **Local Data Sovereignty:** Raw clinical CGM data is never uploaded to remote cloud databases; strictly git-ignored.
- **De-Identification & Zero PHI:** Only anonymized physiological telemetry is processed.
- **Feature Trust Badges:** Every screen explicitly badged ■ WORKING, ■ DEMONSTRATION, or ■ FUTURE.
- **Dual Architecture:** High-performance FastAPI REST API on port 8000 + interactive Streamlit suite on port 8501.